

Using Social Network Analysis in Applied Psychological Research: A Tutorial

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Abstract

Social network analysis (SNA) is a highly flexible research method that allows for novel exploration of a wide variety of research phenomena. Evidence from fields as disparate as public health, education, informatics, sociology, and medicine has demonstrated the importance of recognizing the complexity inherent in individuals' connections with others. In this article, we provide a brief conceptual overview of social network theory and methodology, and then demonstrate how to apply SNA to an applied psychological research context studying students embedded in classrooms. We also provide numerous supporting materials on our OSF page, including R code for all analyses, a dataset containing social network data, and a glossary of key terms in social network analysis. We conclude with a set of recommendations for researchers interested in applying SNA to their own contexts and content areas.

Translational Abstract

Social network analysis (SNA) is a widely used and highly flexible research method that has significant potential for applied psychological work. However, despite the growth in empirical studies using SNA in psychology and related fields, it has less often been the subject of a tutorial. In this article, we provide a brief conceptual overview of social network theory and methodology, and then demonstrate how to apply SNA to an applied psychological research context studying students embedded in classrooms. We also provide supporting materials on our OSF page, including R code for all analyses, a dataset containing social network data, and a glossary of key terms in social network analysis. We conclude with a set of recommendations for researchers interested in applying SNA to their own contexts and content areas.

Keywords: social network analysis, sociometrics, tutorial, R programming

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Social network analysis (SNA) is a highly flexible research method that allows for novel exploration of a wide variety of

psychological phenomena. Although psychological research often focuses on individual outcomes and processes, proximal social influences such as peers, coworkers, partners, and family members shape individual's perceptions of the world, as well as individual actions, beliefs, opinions, and behaviors. Indeed, evidence from fields as disparate as communication, public health, education, informatics, sociology, and medicine has demonstrated the importance of recognizing the complexity inherent in individuals' connections with others. SNA can help psychologists better understand and quantify how connections with others influence individual differences and behaviors within complex social contexts.

Benefits of Using SNA to Answer Psychological Research Questions

SNA is not an entirely novel approach in many psychological disciplines and can be used to answer a wide range of research questions. One common use has been to model the effects of treatment contagion by examining how effects of treatment

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OSF page with supporting materials: <https://osf.io/abkcx/>.

 The data are available at <https://osf.io/abkcx/>

 The experiment materials are available at <https://osf.io/abkcx/>

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exposure may spread through a social network. For example, friends of individuals who receive an alcohol intervention significantly reduce their own drinking (Hallgren et al., 2021) and in the context of an experimental peer task, antisocial behaviors among adults are generally more contagious than prosocial behaviors, particularly for those in close social proximity to peers (Dimant, 2019). SNA has also been used to examine the sociodemographic, behavioral, and personality factors associated with who we choose as friends, and the extent to which those friends influence our beliefs, attitudes, and behavior (McPherson et al., 2001). To illustrate, adolescent peer groups exert considerable influence on individual gender identity, even after controlling for the tendency of individuals to select friends who are already like themselves (Kornienko et al., 2016).

SNA can also be a powerful tool for examining longitudinal data in order to better understand changes in individual behavior and beliefs across time. Past work has spanned a wide range of beliefs and behaviors including studies on language style, smoking, HIV risk perceptions, racial/ethnic identity development, and gang affiliation and prisoner reentry (Burgess-Hull et al., 2018; Gallupe & Gravel, 2018; Koku & Felsher, 2020; Kovacs & Kleinbaum, 2020; Santos et al., 2017). More recently, researchers in the psychological sciences have used SNA to document associations between individuals within social networks and their biological markers

(e.g., hormones, brain connectivity), highlighting the neurobiological underpinnings of human social behaviors (Cheng & Kornienko, 2020; Hyon et al., 2020). For example, a study on hormones and social network dynamics examined friendship ties in a collegiate marching band; findings revealed that individual differences in cortisol and testosterone were associated with the initiation and maintenance of friendships (Kornienko et al., 2016).

SNA methods are also useful in research that examines how individual positionality within a social structure influences access to and distribution of resources and individual behavior (e.g., connections between adolescent sleep insufficiency and popularity; Li et al., 2019). Although not exhaustive, these areas of inquiry and examples help to illustrate that SNA is particularly well suited for psychological research. Integrating individuals and the relationships among them, SNA can answer a range of questions in psychological research on location, membership, and attributes of an individual within a network, as well as social relations (e.g., affect, kinship), interactions, and transfers of knowledge or resources between individuals (Borgatti et al., 2009). Thus, SNA offers value to psychological researchers by opening access to a diverse set of empirical questions and methodologies (Clifton & Webster, 2017).

SNA has also shown promise and potential in many applied psychological subfields and settings with rich embedded social systems (e.g., schools, community centers). For example, SNA has been used in education and educational psychology to analyze college student retention and how social interactions impact the decision to leave or stay enrolled in higher education (Eckles & Stradley, 2012). It has also been used to understand interaction patterns in e-learning environments (Cela et al., 2015), and to analyze extracurricular involvement and friendships (Schaefer et al., 2011). Outside of education, studies also used SNA to examine leadership within college sports teams (Loughead et al., 2016) and the influence of participation in college sports on psychological

wellbeing (Graupensperger et al., 2020). In the mental health literature, SNA has been used to examine the role interconnectedness plays within a community at risk for depression and posttraumatic stress disorder after a disaster (Bryant et al., 2017) and the influence of perceptions of peers' drinking behavior on college students' own drinking behavior (Cox et al., 2019). It has also been used to compare demographic variables such as differences in social well-being by age (Bruine de Bruin et al., 2020) as well as racial-ethnic differences of perceived support (Flores et al., 2020), and has been applied as a psychometric tool useful for conceptualizing and visualizing alternative factorial structures for mental health conditions (Massidda et al., 2016).¹

Given the breadth of purposes and uses across multiple content domains, SNA is a flexible and widely applicable analytic approach that can provide new insight into complex psychological phenomena. Like other broadly applicable methods such as multi-level modeling, machine learning, and meta-analysis, SNA offers a rigorous pathway to deeper scientific understanding. However, to our knowledge, SNA has primarily been featured in the context of empirical research but is much less common as the focus of a methods tutorial in psychological journals.

Purpose of Tutorial and Intended Audience

Given the potential of SNA as a research framework for understanding and modeling social complexity, and the relative sparsity of SNA tutorials in applied psychology, the purpose of this article is to demonstrate how SNA can be applied in a field setting (in this case, classrooms) to better understand the interactions between psychological and cognitive constructs and the surrounding social space. We recognize that SNA encompasses a vast array of methods, techniques, and theoretical approaches, such that an exhaustive treatment of all would be impossible in this format. However, we believe that the approaches we describe here are of particular interest to psychological researchers who work in applied settings with rich embedded social systems (e.g., schools, organizations, workplaces) and wish to better understand how individuals' psychological and cognitive dimensions may be shaped by, or may shape, those of their proximal peers, as well as how to marry this conceptual understanding with corresponding statistical methods.

Foundations of SNA: The Work of Jacob Moreno

Many of the foundations of SNA, including data collection methods, visualization techniques, and sociometric mathematical modeling, were introduced in the 1930s through the work of psychiatrist Jacob Moreno (Freeman, 2004). Through his research, Moreno highlighted the potential of SNA to better grasp the complexities of interrelations within a population. His interest in social structure was first evident in his work *Das Stegreiftheater* (Freeman, 2004). This work drew on the philosophies of psychodrama and sociodrama as a therapeutic technique and analyzed the interactions among actors. Moreno and a fellow researcher, Helen Hall Jennings, continued this work at Sing Prison (Moreno, 1932) and

¹ Table A1, available in the online appendix via our Open Science Framework (OSF) site at (<https://osf.io/abkcx/>), includes more detailed descriptions of each of these studies, including summaries of the research purpose and network analysis approach that was used.

the Hudson School for Girls (Moreno, 1934). These studies included systematic observation and analysis of social networks, and established a foundation for the sociometric approach, using the term “network” as we know it today. Moreno, Jennings, and mathematician Paul Lazarsfeld also developed the first mathematical models to analyze social networks and created a model for sociometric choice (Moreno & Jennings, 1938).

Sociometric Versus Egocentric Network Research

Since Moreno’s pioneering work, the field of SNA has expanded steadily, encompassing many distinct methodologies and theoretical paradigms. One of the **key distinctions when planning to conduct social network research is whether to take a sociometric or egocentric perspective** (Wasserman & Faust, 1994). Research from a **sociometric** or general network perspective seeks to **understand entire intact networks** within existing social spaces (Butts, 2008b). The goal here is to capture as much information as possible about the existing connections in the social space of interest (e.g., classroom, workplace). Thus, one might ask participants to name their closest colleagues, friends, and so forth, and ask them to evaluate these connections according to some qualitative distinction, such as quality of support, information provided, resources, and so forth. This establishes a baseline network of nominators (also known as egos) and nominees (also known as alters) and denotes the strength or quality of ties between each actor.

Sociometric research typically requires one to bound the network(s) of interest in specific ways. In the example we use for this tutorial, students in kindergarten classrooms were asked to identify their *three closest friends in their class*. This last part is important because it bounds the network to only students who are in one classroom (and bounds the number of possible outgoing connections for a given student to three). Friends from the neighborhood who go to different schools or friends from different classes or grades would not be reflected using this perspective. Thus, any questions being pursued using a sociometric paradigm must also be conceptually and theoretically bounded to the specific network in question (e.g., athletes on a sports team, marching band members, sorority members, mental health providers within the same clinic, nursing staff on a unit, juveniles within a center). Further, this approach requires the researcher to have access to a class roster, organizational chart, membership list, or some similar method to link participants to each other within the same network.

In contrast to sociometric research, **egocentric network research seeks to understand the structural and functional variation within individuals’ social networks but does not bound the network to a particular social or physical space**. Each individual network is treated separately, and these individual networks are typically compared in terms of their structure and composition. Often, researchers will also collect “second degree” tie information from each participant (e.g., they may ask “of the people you nominated as friends, who knows each other?”). This allows for estimation of additional network statistics such as the density of ties between alters nominated by a given ego. A key distinction between egocentric and sociometric research is that unlike in sociometric research, egocentric methods do not typically require researchers to begin with a roster or other complete list of potential nominees. Thus, it is not bound to a particular social or organizational space,

and rather is conceptually bound by the individual’s perceptions of their network. But it does not allow for a deeper investigation of some key network properties, such as centrality (see below), that are possible using a sociometric research approach.

Both paradigms are useful and important for a better understanding of how we are shaped by social context and connections. In this tutorial, we will adopt a sociometric perspective, as we believe it often aligns more closely with the conceptual framework and underlying questions that motivate many psychological research questions. McCarty et al. (2019) offer a thorough introduction to egocentric methods for those interested in this approach.

Outline of the Tutorial

In the following tutorial, we begin with a brief overview of key terminology within social network research. We then discuss the most effective ways to collect sociometric research data in applied settings. Next, we describe how network data collected via surveys can be converted into data structures most useful for conducting SNA. We briefly describe how to visualize social networks as well as how to use network data to estimate measures of centrality for each individual. Finally, we discuss how to use SNA data to fit statistical models of influence and selection, two of the most common approaches used to understand how individuals impact and associate with one another in a network. We conclude with an applied example from Chow et al. (2021), in which students’ language skills are used as predictors of an individual student’s centrality to their classroom network. Additional resources and practical guidance, including a glossary of key terms in SNA, as well as R scripts and a dataset with network data, are available as part of an online appendix to this article found at our Open Science Framework page (<https://osf.io/abkcx/>).

Overview of Key Terms and Concepts in SNA

SNA operationalizes a set of connections between actors in a network. A wide range of social units can serve as actors in a network, including but not limited to individual persons, organizations, groups, teams, and nations. Actors are also referred to as nodes, egos, or vertices. Similarly, a wide range of connections can be represented between actors, including affective ties (e.g., friendships, supportive colleagues), physical proximity (e.g., who lives by who), interactions (e.g., drug sharing patterns), and relationships (e.g., marriages, sexual partners). Connections between actors are also referred to as ties, links, arcs, or edges. The connections between actors can occur in a bounded or unbounded network. A bounded network consists of a set number of actors that is known by the researcher prior to collecting data (e.g., students in a classroom). An unbounded network has an unlimited set of actors (e.g., all the people who might support a college student with mental health or academic help). These two network structures yield different levels of information; bounded networks provide the greatest level of detail for social network analysis and depicting social structures within the network (Wasserman & Faust, 1994). When choosing between a bounded or unbounded network, it is important to carefully consider research questions and study design.

Ties between two actors can be directed or undirected. Directed ties depict connections between two actors that may not be reciprocal. For example, in a friendship network, Actor A nominates

Actor B as a friend, but the nomination is not reciprocated. In graphical form, the direction of this tie can be depicted as a one-way arrow stemming from Actor A (the sender of the nomination) and connecting to Actor B (the receiver of the nomination). It is important to note that a directed tie can also be reciprocal. For example, both Actor A and Actor B can nominate one another as friends. In graphical form, this tie can be depicted as a bidirectional arrow connecting Actor A and Actor B. The bidirectional arrow indicates that the actors in this dyad are both nomination senders and receivers. In contrast, undirected ties do not specify which actors are senders and which are receivers. An undirected tie simply indicates a connection is present between the dyad.

Ties can also depict the strength of a connection between actors. This is commonly referred to as weighting the connection between two actors. For example, a friendship network may assign a value to the connection between actors based on a frequency count of how often actors report socializing with one another. In contrast to a binary network, weighted connections are referred to as valued networks, in which each dyad is assigned a value to depict the intensity of the connection between actors. Another example of weighting in a network might be to assign values to ties between two actors according to the number of shared needles in a network of injection drug users (Unger et al., 2006).

Often, connections between actors are modeled as one-mode networks, in which all the actors in a network represent the same unit (e.g., individuals, organizations, nations). Networks can also be modeled as two-mode or bipartite networks. Two mode networks have two distinct types of actors. In these models, the first type of actor (e.g., individuals) reports on their ties or membership to a second type of actor (e.g., attendance at social events, membership in university clubs). To ease interpretability and to analyze data in SNA frameworks, two mode networks are often transformed (i.e., projected) into one-mode networks (Opsahl, 2013). Projection is accomplished by selecting one of the sets of nodes and linking nodes from that set. To illustrate, in a network describing observations of play during center time at a preschool, if Student A and Student B were both observed at the block center, Student A and Student B (first level actors) would be connected because they have a common tie to the block center (second level actor). These networks can be analyzed in a binary approach; however, they can also be weighted. For example, the tie between Student A and Student B could be weighted by the total number of centers that they both visited.

An additional consideration for defining a network is multiplexity. A connection between two actors is multiplex when the actors interact across multiple different types of relations (Butts, 2008b). To illustrate, the connection between Student A and Student B may be multiplex because the students are in the same classroom, however, these students also interact outside of the classroom because they are neighbors. Depending on the research question, the greater the degree of multiplexity between actors in a network, the greater the risk of skewing connections between network features and outcomes of interest. However, in psychological sciences, researchers may be interested in how multiplex relationships function as part of an empirical research question. For example, one might examine how the overlap, shape, or structure of resource sharing or treatment contagion is influenced by multiplexity in a network, as found in Felsher and Koku (2018), who examined multiplexity across overlap in drug-use, needle exchange, and sexual relations as

a risk factor for HIV. Another common focus of research using network multiplexity is to compare individuals' "online" and "offline" relationships or behaviors (e.g., Mesch & Talmud, 2006) to quantify their relative impact and/or influence on each other.

In addition to mode and structure, networks can also be defined by the sampling method. Complete networks aim to collect tie information from every actor in the network (e.g., asking each patient in a treatment facility about their friendship ties within the facility). This approach is time consuming and often difficult to accomplish in authentic settings such as workplaces or classrooms. Other methods are less time intensive and require participation from fewer actors. However, these methods may yield less information about the network and structures within. For example, one might randomly sample a subset of individuals within a given network (e.g., several students within a classroom), and then use that subsample to generalize to the larger network (e.g., all students in the classroom), or one might ask a key informant, such as a teacher, to report on connections between their students, as described in more detail below.

Collecting Sociometric Network Data

When collecting sociometric data, researchers must first decide on what kind of network as well as what kind of connection/relationship they will study as these decisions will help determine the choice between four main data collection methods. Network data can be collected through interviews or surveys (Scott & Carrington, 2012), observation (Gibson, 2005), archives and historical materials (Gould, 1995; Padgett & Ansell, 1993), or from trace observation of electronic communications (Carley, 2006). Scott (2017) provides an excellent overview of all four of these methods.

The most common approaches in social science are interviews and surveys. When conducting a survey of an entire network, researchers ask respondents to report on their relations with other individuals within the network. This typically involves giving participants a list of other members in their network and asking them to note with whom they share a tie (e.g., a researcher may present a roster of classmates and then ask a participant "Who do you hang out with most?"; Butts, 2008b). If no such list is available or it is too large (e.g., in an organization with thousands of members), respondents can create their own list of relations, recalling a specified number of people with whom they have a relevant connection (e.g., a researcher may ask "Thinking about all the students at your school, who do you hang out with most?"). Information regarding the importance or strength of a relation can be gained through follow-up questions (e.g., a researcher may ask "Thinking about the student you just named, how would you rate the quality of your relationship?"; Scott, 2017). Egocentric network data are commonly collected using name generators. Name generators allow participants to list the people they share a particular relation with (Burt, 1984; Hogan et al., 2007; Marsden, 1987, 2005) and can be followed by name interpreter questions. These questions ask about attributes of alters or relationships noted in the name generator (Scott, 2017), such as the nominee's relationship to the nominator (e.g., parent, friend, roommate, teacher, counselor).

Unstructured or semistructured interviews and open-ended questions are typically used for interview-based data collection methods. Interviews ask similar questions to those presented in a

survey, but here a researcher walks the participant through the questions and can probe for more information based on individual responses. In comparison to a survey format, an open-ended interview approach can be beneficial for participants who may find a survey too overwhelming and/or impersonal, for participants who need additional guidance due to the complexity of the specified relationships, or for young participants who may need additional assistance (Baumgarten & Lahusen, 2006; Franke & Wald, 2006).

Relatively recent technological developments have also improved the survey and interview experience. For example, electronic network data collection can reduce nonresponse and attrition due to respondent fatigue (Bowling, 2005 as cited in Adams, 2020). Apps and online platforms have also been incorporated into data collection methods (Adams, 2020). For example, interactive apps have been developed as a method for gathering personal network information (Hogan et al., 2007). These are typically implemented using a tablet, smart phone, or laptop, and involve building a personal network step by step as participants add in new information (e.g., EgoWeb 2.0, n.d.; Network Canvas, n.d.).

Observations can also be utilized in a more ethnographic approach to SNA. Observations are usually not a primary data source and are typically complemented by interviews or surveys (Scott, 2017). Via observation, the researcher can observe participants' natural interactions with one another, such as colleagues having conversations in the workplace or children playing together at recess. To illustrate, Fabes et al.'s (2003) observations of young children's play quality revealed children rarely played only with other-sex peers, but approximately one fourth of their interactions involved play with mixed-sex peers. These interactions can then be coded and converted into data suitable for network analysis. Archival data and documents such as journal articles, letters, emails, blogs, tweets, and so forth, can also be useful in generating data for SNA. Connections between individuals can be made by tracking emails sent and received, tweets and other social media posts that are "Liked", favorited, or mentioned, or coauthors that are listed in a publication. These data are advantageous in that they are relatively easy to access and often free or low cost.

Another important factor to consider when collecting social network data is the timing of data collection. Longitudinal data can provide a greater level of detail and insight to social networks and underlying structures and processes compared to cross-sectional data. If multiple waves of network data are collected from the same participants, researchers can then assess patterns of stability and/or change in nominations over time (Carolan, 2014), as well as measure associated outcomes that may be related to those social connections. To illustrate, results across nine years of the Mind Your Heart study showed social network size and greater perceived social support predicted lower suicidal ideation chronicity and severity among veterans (Arenson et al., 2021).

Handling Missing Network Data

Missingness is a key consideration when designing studies using SNA. In some ways, the problems associated with missingness are compounded when collecting sociometric data. For example, suppose in a given classroom ($N = 20$ students), five students were absent on the day(s) that network interviews were conducted, rendering their classroom tie information missing. This is problematic because the study is now missing data on 25% of the network actors and more than 40% of the potential network ties, which may significantly

influence the interpretations of these data. To further complicate this, even while absent, those five absent students may have been named as alters by the students who were interviewed. This leaves the researcher in a difficult situation to include only the incoming ties to those absent students, or to remove them entirely from the network.

This issue can be even more challenging in the situation where students are not able to participate in the research, perhaps because parental consent was not provided for a younger student, or because an older middle or high school student does not assent to participate.² This would mean that even if a student received nominations from their classmates, they could not be included in the network data in any form (absent consent and/or assent). Thus, the network again is subject to bias due to missing or unavailable data on ties and alters. As a rule, in school-based research, at least 60% of network members must be included to provide reasonable assurance that a stable representation of the complete network has been captured (Huisman, 2009). However, as more recent work has demonstrated (e.g., Smith et al., 2022) relying only on this rule of thumb is not recommended, as it often overlooks important considerations that will have a direct impact on the amount of bias that may be introduced due to missing data, such as the type of network being analyzed (e.g., directed vs. undirected), and the specific network measure of interest (e.g., degree centrality vs. closeness). Thus, a more in-depth examination of missingness is usually required.

If researchers are concerned about the rate of missingness in their network data, there are alternative approaches that can help to resolve this issue. For example, one might ask all or many network members about all potential network connections, thereby mapping cognitive social structures (Krackhardt, 1987; Neal, 2008), which can then be used to determine whether a tie may be present for an absent student. Analytically, one can also use a combination of multiple imputation (Jorgensen et al., 2018), often in concert with simulation data from exponential random graph models (ERGMs; Krause et al., 2018), to simulate a range of possible plausible network connections for a missing participant, and then incorporate this uncertainty into any subsequent statistical modeling (see below). But in general, working with missing data in networks is an active area of methodological development, and researchers are encouraged to avoid missingness with careful initial planning. For more specific guidance, Smith et al. (2022) provide a useful flowchart for identifying the best approach to use for handling missing data depending on the specific network type and measure of interest being used.

Transforming Survey Data Into Network Data

After collecting network data, the next step is to convert the data into a structure that allows for network analysis. We describe two of the most common structures that are also easily converted into network data using the *igraph* package in R (Csardi & Nepusz, 2006), edgelist, and adjacency matrices. An edgelist is a two-column data frame where each row represents a unique tie. Table 1 provides an example of this data structure. Here, we see that Student A nominated Student B and Student C as classroom friends. Student B nominated Student A, and Student C nominated

² We recognize that this is a situation that may be unique to research with children, which when collecting network data typically requires some form of parental consent. In an applied research setting where the researcher does not have to obtain consent from an outside party, this issue may be somewhat less critical.

Table 1
Example of Edgelist Data Structure

Nominator (ego)	Nominee (alter)	Frequency
A	B	10
A	C	5
B	A	1
C	A	7
C	B	4

Note. Frequency here refers to the number of times the nominator reported having a conversation with the nominee longer than 1 min in the past week.

Student A and Student B. If students also provided information about the quality or strength of each tie (e.g., how often they work together), this tie-level information can be easily incorporated with additional columns to note these values in the edgelist. In [Table 1](#), for example, a variable named “Frequency” is included, which represents the number of times in the past week the nominator reported having a conversation longer than 1 minute with the nominee. Creating an edgelist from traditional “wide” survey data structure (where one row represents one student, and multiple columns represent ties/nominations), one would need to reshape their data from wide to long format. For more detail, we demonstrate how to apply this transformation in the R code referenced above.

An adjacency matrix represents network ties in matrix format, where all actors are listed in both the rows and columns, and ties are represented as 1’s in a particular cell ([Butts, 2008b](#)). This data structure is less common when collecting directed ties but works well with undirected ties. [Table 2](#) below is an example of an adjacency matrix for the same set of three students. Here, we see Students A–C listed in both the columns and the rows. If a tie is present between two students, a “1” is present, otherwise “0.” This approach can also be extended to value or weighted networks when collecting information about the quality of ties. This is accomplished by replacing “1” with the intensity or other quality being measured (e.g., frequency). We have found that the adjacency matrix is a compact and relatively easy to use structure for collecting data from key informants (e.g., teachers reporting on all friendships within their classrooms). Both these data structures allow for the collection of both weighted and unweighted information about network ties. However, when there are a large number of nodes, the adjacency matrix can become unwieldy, as it is hard to evaluate a large $N \times N$ matrix. An edgelist is especially effective for a compact display of all ties in a network, especially when the network is sparse.

Table 2
Example of Adjacency Matrix Data Structure

Student	Student A	Student B	Student C
Student A		1	1
Student B	1		1
Student C	1	0	

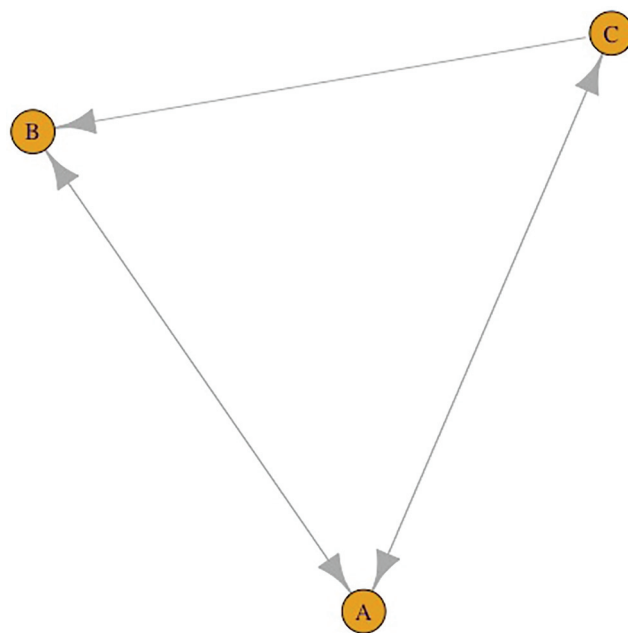
Note. Frequency here refers to the number of times the nominator reported having a conversation with the nominee longer than 1 min in the past week.

Visualizing Network Data Structures

Once network data have been converted to a data structure conducive to network analysis, a typical first step is to visualize the network using a sociogram. A sociogram is a visualization that represents the nodes (actors) and edges (ties) of a network as a series of connected points. Sociograms can efficiently present a large amount of information about a given network and can be used to visually identify network factors such as cliques or subgroups, actors who serve as bridges between peer groups, and the density of network ties. Arrows can be used to represent directed ties, and weighting can be used to reflect measures of quality about each tie (e.g., thicker lines suggest higher frequency of contact or more support). In addition, nodes can take on different shapes, colors, or sizes to reflect different characteristics of the actor (e.g., race/ethnicity, gender, importance in the network), or to denote discrete subgroups or cliques within the larger network. [Figures 1](#) and [2](#) are sociograms derived from the dummy data from [Table 1](#). [Figure 1](#) is an unweighted, directed graph. Double-headed arrows represent reciprocal relationships, and single-headed arrows represent unidirectional relationships (i.e., Person C nominated Person B, but Person B did not nominate Person C). [Figure 2](#), in contrast, is a weighted, undirected graph. Here, the thickness of each line represents the frequency of interaction, as measured by the frequency variable in [Table 1](#). Edges between nodes here represent any nomination, regardless of direction.

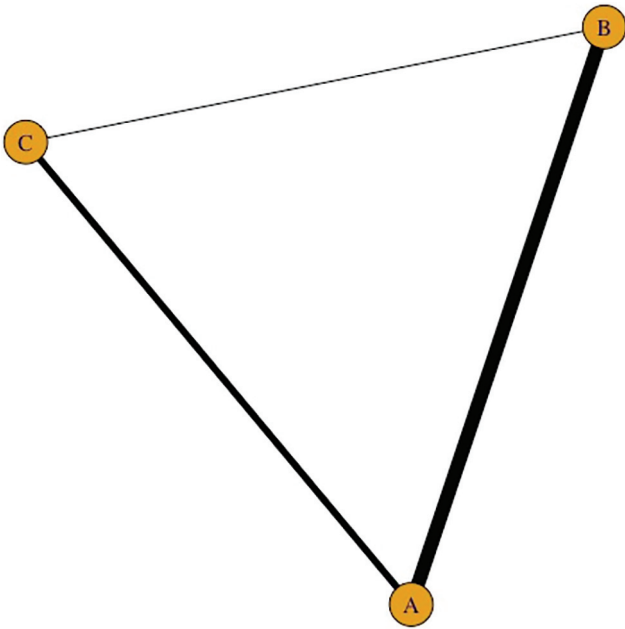
[Figure 3](#) is a composite of four classroom sociograms created using data on classroom friendships found in our example dataset. Here, we see the friendship network from Classroom 101. Arrows are used to represent the direction of ties, so a tie with arrows in both directions suggests a reciprocal tie between students. Triangles (as opposed to circles) are used to denote students who were

Figure 1
Directed, Unweighted Sociogram Based on the Data From Table 1



Note. Nodes (people) are represented as circles. Edges are represented as arrows, which also indicate the direction of each friendship nomination.

Figure 2
Undirected, Weighted Sociogram Based on the Data From Table 1



Note. Nodes (people) are represented as circles. Edges are represented as line segments, which are weighted according to frequency of interaction, with thicker lines indicating more interaction.

determined to be at risk of specific language impairment (SLI). This allows the reader to visualize the relationship between an actor's position in the network and their risk status, a central question in the study. The igraph package provides numerous different layouts for social network data, including specific algorithms to determine the placement of each actor on the grid space.

Calculating Network- and Actor-Level Summary Statistics

Visual inspection of social networks can often provide useful insight into the structure and function of social spaces. However, especially when dealing with multiple intact networks that each could have many actors, it is useful to calculate network- and actor-level statistics to summarize the properties of each network and describe each individual actor's contribution to network dynamics.

In this section, we draw from Wasserman and Faust (1994) to provide an overview of terminology and definitions of network size, density, reciprocity, transitivity, diameter and distance, clustering, and centralization. Network size is measured by counting the number of nodes in a network. Density is the percentage of possible links that are present in the network and is given by the following formula:

$$D = \frac{T}{N(N-1)}$$

where N is the network size and T is the total number of ties in the network. More dense networks are characterized by having more connections between nodes, with 0 representing a network with no

connections, and 1 representing a network where all nodes are connected to each other.

Reciprocity, also referred to as mutuality, is defined as the degree to which actors in a network select one another mutually. It analyzes the extent to which relationships in a network are reciprocal. Reciprocity can indicate the direction of resources as well as sources of stability in a network. Estimates of reciprocity can also indicate whether networks are equal (high levels of reciprocity) or hierarchical (low levels of reciprocity). For example, in a typical workplace, managers with more "top down" leadership approaches may often convey information to their employees, but employees may not select them as an important or influential colleague. This could lead to a hierarchical network with lower levels of reciprocity. Workplaces with more bidirectional information flows between managers and subordinates might also have higher levels of reciprocity in the network.

Diameter is the longest path, or "sequence of distinct, serially adjacent vertices" (Butts, 2008b, p. 15) between two actors. This is an important distinguishing network measure in that networks can have the same size or density, but different diameters. Distance is another network level property that analyzes how far actors are from one another. Distance measures the average path length which measures the mean distance between all pairs of actors in the network. This can help indicate the cohesiveness of the network. Diameter and distance are closely related in that the characteristics are similar in complete networks but are calculated independently in ego networks. These can be applied to indicate how efficiently resources, norms, or information can move from one part of the network to another.

Transitivity of a network reflects the social structure's tendency toward stability and consistency. This examines triads, or groups of three actors. For example, in a classroom network, if Student A collaborates with Student B, and Student B also collaborates with Student C, then one might assume that Student A collaborates with Student C through their transitive relationship with Student B. Triadic relations have four possible types. This includes no ties, one tie, two ties, or all three ties. One can summarize the overall transitivity of a network by summarizing the level of connection among each potential triad in the data. In this example, transitivity is defined for undirected ties, but transitivity can also be extended to directed networks (Wasserman & Faust, 1994).

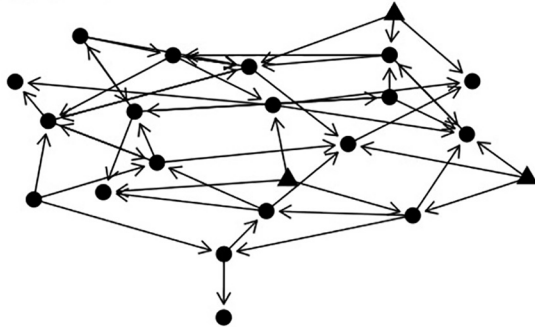
For our work with students in classrooms, we tend to focus on actor-level statistics, which often map more closely onto the questions we want to answer about how students from different groups and with different characteristics function in relation to the social space of the classroom. The statistic we find most helpful for these questions is degree centrality, along with two related statistics, betweenness and closeness (Freeman, 1979; Marsden, 2002).

Degree centrality is the sum of all ties, both incoming and outgoing, for a given actor in a network. For example, if a student nominates three different friends (outgoing ties) and is nominated by six other classmates (incoming ties), that student would have a centrality score of 9, the sum of outgoing and incoming ties. When comparing multiple networks of varying size, as we often do with classrooms, it is recommended to normalize centrality scores by dividing each score by $(N - 1)$, where N is the size of each classroom network. Centrality scores can also be separated into in-degree and out-degree scores, especially in cases where one might

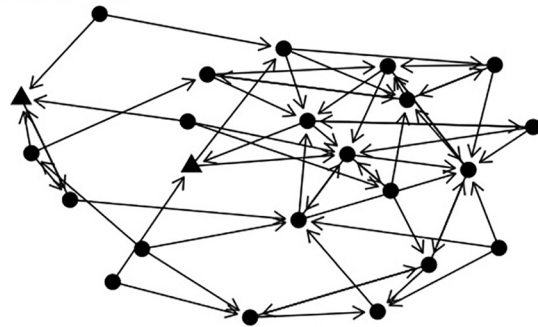
Figure 3
Sample Network Diagrams From Example Data

Classroom Friendship Networks
 by Student Risk Status

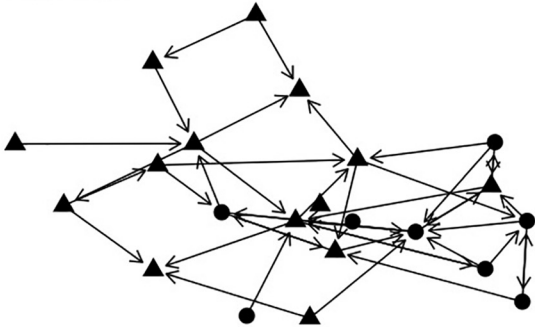
Classroom 101



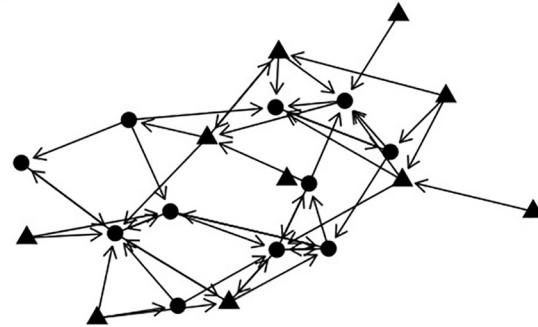
Classroom 103



Classroom 201



Classroom 401



Note. Students at risk of language disorder are denoted with triangles. Students not at risk are denoted with circles. Arrows indicated the direction of each friendship nomination.

have specific hypotheses about differences between sending versus receiving nominations.

Closeness is a centrality measure that accounts for the pattern of ties in an actor's network. This looks at the relations between the alters of an actor (i.e., friends of friends) to capture the average distance an actor is from all other actors. This is calculated by first summing the geodesic distance of the chosen actor and all other actors in the network. This result is then inverted and then multiplied by $(N - 1)$, where N is equal to the total number of actors in the network.

Betweenness examines how actors mediate relationships between other actors that are not directly connected. This is important to look at when measuring exchanges of information or resources within a network (Knoke & Yang, 2008). Intermediary actors between exchanges are considered more influential the more they are used. This measure also emphasizes measuring the degree actors fall on the shortest path between pairs in a network. Betweenness is calculated by adding the number of times an actor falls on the shortest path connecting others in the network.

Fitting Influence and Selection Models With Network Data

As seen above, social network data can provide a wide range of summary statistics and visualizations that can be used to answer

research questions about group formation, friendships, flows of information and resources, and much more. However, another strength of SNA is that it translates readily into several statistical modeling frameworks. As Frank (1998) outlines, there are two general "classes" of models that are commonly used with individual longitudinal network data. Influence models seek to understand how one's closest relationships (family, friendships, colleagues, neighbors, etc.) might influence an individual actor's attitudes, beliefs, or behavior. The concept of "peer pressure," though somewhat different than what is described here, may be a useful heuristic. Often, people are shaped by those closest to them, and influence models attempt to quantify the magnitude of this influence.

Influence models are typically estimated using network autocorrelation models (Doreian, 1981; Ord, 1975). These models operate under the assumption that the beliefs, actions, decisions, and so forth of actors in the same network are correlated, and the strength of that correlation can be estimated with a network autocorrelation parameter (typically represented as ρ). This parameter serves as a test of the influence mechanism of the network being examined. Originally developed for the analysis of first-order network data with information on only one network process, the model has since been extended to higher-order relationships (Elhorst et al., 2012), and most recently, can now be implemented in a Bayesian framework that allows for considerably more flexibility in model specification and hypothesis testing as to the relative strength of different

social network mechanisms and/or processes (Dittrich, 2020). For example, one might examine teachers' advice seeking and information sharing networks concurrently and examine their collective and/or relative influence on their sense of teaching self-concept.

First-order influence models can also be estimated in a multilevel modeling framework, with individuals nested within larger networks. These may be intact networks, such as students nested in classrooms, or in some cases, individuals are nested within their local positions (subgroups or cliques, Frank et al., 2013). As with all multilevel models, covariates can be introduced at both the individual and network levels, allowing for highly flexible model specification and the inclusion of several different possible confounding variables. When estimating a full influence model, individual prior behavior, beliefs, or outcomes are usually also included to obtain more precise and unbiased estimates of the group of network influence effects.

If longitudinal network data are collected, several models can be used to assess the strength of influence mechanisms, including separable temporal exponential random graph models (STERGMS; Krivitsky & Handcock, 2014), stochastic actor-oriented models (SAOMs; Snijders, 1996, 2001). These models can simultaneously account for changes in the underlying network structure (e.g., adding or removing ties between nodes) while controlling for covariates at the node or dyad level. Network change, or evolution, can then be modeled as functions of these characteristics, which in turn can be treated as random or fixed depending on the network properties of interest and the available data.

Relational event models (Butts, 2008a) are another highly flexible framework for modeling longitudinal network data, particularly when the data describe a series of social "events" or interactions over time between a social actor ("sender") and a set of "receivers," which might be other actors, locations, or combinations of both. These models can be parameterized to describe a number of characteristics of the system being studied, including individual heterogeneity in sending and/or receiving, the tendency for certain actors to be more attractive to senders, and the presence of triadic relationships, where two actors are influenced by interaction with a third party (Butts, 2008a).

For example, DuBois et al. (2013) used data from direct observation of 278 high school classroom sessions to estimate a series of relational event models that described the likelihood of students interacting with one another during class. The models included information about classroom seating, the style of teaching being used, students' gender, grades, and the classroom context, such as whether students were listening to lectures, doing group work, and so forth, as well as recency effects (whether two actors had previously interacted, and how recently), and dyadic covariates, such as the number of school activities two students had in common. Thus, relational event models offer many avenues to incorporate variables that attend to social and situational context to more precisely model complex social processes over time.

Along with influence models, selection models (Wasserman & Faust, 1994) are a dominant modeling approach. They are in some ways seen as the inverse to influence processes. While influence modeling seeks to understand how those closest to us influence our thoughts or behaviors, selection modeling seeks to understand how we come to be close to some people and not others. In other words, selection modeling aims to quantify the factors that may impact whether a given actor nominates a particular alter as a

friend or colleague. These processes typically are seen as complementary, and often cyclical - we choose our friends based on given characteristics, criteria, or shared context, and then those friends in turn shape how we think, feel, and act. Over time, this cycle is repeated.

The most common approach to model selection is the ERGM, or p* model (Frank & Strauss, 1986; Snijders et al., 2006; Wasserman & Pattison, 1996). ERGMs are highly flexible models that use Markov Chain Monte Carlo (MCMC) maximum likelihood estimation to test specific hypotheses about network nodes, dyads, or complete networks. Luke (2015) describes ERGMs as a "generative" modeling approach, as they allow researchers to test hypotheses about individual nodes (or dyads) and examine their impact on the larger structural and compositional features of a network.

Another common approach used to model selection is the "p2" model (van Duijn et al., 2004; Zijlstra, 2006), which models the likelihood of individual dyads being formed while also controlling for covariates related to the nominator (sender) and the nominee (receiver). This is an area of active methodological development, as Sweet (2016) demonstrates. Latent space models (Hoff et al., 2002) and actor-oriented stochastic block models (Niezink et al., 2019) have been shown to model network data more flexibly, and in some cases with longitudinal data, model the coevolution of influence and selection processes over time.

A Pedagogical Example

Having discussed many of the elements of conceptualization, design, and analysis of research using social network data, we now provide one example of recent work that applied these elements to classroom research. This example includes data from a study that hypothesized language skills would play an important role in early social settings, particularly because communication is key in forming relationships and understanding peers as well as broader social dynamics. In this tutorial, we include a subset of data from this larger study (currently under review). We were particularly interested in understanding the role that language would play as an antecedent of centrality. Although conducted in a different setting, the overall structure of this work is like that of Klein et al. (2004), who sought to understand the antecedents of centrality in a workplace/organizational context.

Data Provided for the Tutorial

We recognize that pedagogical examples often are most useful when sample data are provided. So, we include a dataset here that combines measures of language skills and several other background and demographic variables, along with friendship nominations (the social network component) for a subsample of 188 students nested within nine kindergarten classrooms. The data are available along with code in R for all analyses and visualizations via our Open Science Framework site at (<https://osf.io/abkcx/>).

The example dataset, langnet.dta, is in Stata format, but Stata is not required to open or use the data. The example code includes a section on importing the data into R using the `read_dta` function from the `haven` package (Wickham & Miller, 2020). The dataset includes eleven variables. `Student_id` and `teachid` are unique student and teacher identifiers, respectively. `bf1`, `bf2`, and `bf3` are friendship

nominations for each student. The nominations are linked to student_id to facilitate easy creation of an edgelist. Two measures of students' language are included. Celf_score includes each student's score on the screening tool of the Clinical Evaluation of Language Fundamentals–5th Edition (CELF-5; Wiig et al., 2013) collected at the beginning of the year. Risk_status is a binary variable equal to one if the child scored above the clinical threshold for a risk of a language disorder, and 0 if the student scored below the threshold of risk. We also include several background and demographic variables, including students' age in years (age_yrs), gender (gen), race/ethnicity (race_eth), and the size of their classroom (csize). For reference, Table 3 includes descriptive characteristics of the sample, separated by language risk status.

Step 1: Collecting Network Data and Creating Network Objects

Network data were collected from kindergarten students via individual interviews with research assistants in late Fall and early Winter of the school year. This provided sufficient time for students to adjust to the classroom and their new peers, as well as sufficient exposure to their peers to develop preferred interaction partners. Children were asked who their three best friends in their classroom were. Interviewers had classroom rosters (which represent the bounded classroom network) and provided redirection if children named individuals outside the classroom.

The variables bf1–bf3 referenced above contain the network data on best friends collected from students. The first step is to take this raw data and convert it into an edgelist, a data format conducive to analysis with the igraph package in R. This involves using the select function from the dplyr package (Wickham et al., 2020), and then transforming the data from wide format (one row per student interviewed) to long format (one friendship tie per row). This edgelist containing only ties can then be converted into a network object using the graph_from_data_frame function in igraph. We retain the other information about students (e.g., demographics, risk status) and store it as information about the nodes. These node attributes can then be incorporated into plots, graphs, and other summaries.

Table 3
Descriptive Characteristics of the Sample by Language Risk Status

Variable	Not at risk (N = 112)	At risk (N = 76)	Overall (N = 188)
Age (years)			
M (SD)	6.77 (0.318)	6.88 (0.340)	6.81 (0.330)
Mdn [Min, Max]	6.73 [6.25, 7.65]	6.91 [6.26, 7.49]	6.85 [6.25, 7.65]
Race/ethnicity			
White	80 (71.4%)	40 (52.6%)	120 (63.8%)
African American	16 (14.3%)	12 (15.8%)	28 (14.9%)
Asian	2 (1.8%)	2 (2.6%)	4 (2.1%)
Hispanic/Latinx	2 (1.8%)	11 (14.5%)	13 (6.9%)
Other	10 (8.9%)	8 (10.5%)	18 (9.6%)
Missing	2 (1.8%)	3 (3.9%)	5 (2.7%)
Gender			
Female	53 (47.3%)	32 (42.1%)	85 (45.2%)
Male	59 (52.7%)	44 (57.9%)	103 (54.8%)
Female	53 (47.3%)	32 (42.1%)	85 (45.2%)
CELF score			
M (SD)	11.2 (2.29)	7.01 (2.09)	9.53 (3.03)
Mdn [Min, Max]	11.0 [5.00, 18.0]	7.00 [2.00, 13.0]	9.00 [2.00, 18.0]

Note. CELF = clinical evaluation of language fundamentals.

Once you have created a network object in igraph, you can obtain quite a bit of useful information about the network by printing the name of the object to the console (e.g., typing "g0101" for classroom 101). "D" indicates that the graph is directed, and "N" indicates that the vertices (nodes) of the graph are named (in this case, named with five-digit student identifiers). On the same line, we then see the total number of vertices (22 total in classroom 101) and edges (49). On the second line, we see that several attributes have been added to the network, including vertex names (name) and each student's risk status (risk_status). Finally, we see a listing of all the edges found in this network, with arrows indicating the direction of the friendship nomination (e.g., Student 10108 nominated Student 10120 as a friend). Entering? igraph::summary.igraph into the R console will load the help file for the igraph summary, which provides additional details.

Step 2: Describing Classroom Networks Statistically and Graphically

Both graphical and numerical representations of classroom networks are useful to provide and simple to generate with the igraph package. An extension to igraph, the ggraph package provides an intuitive system for plotting nodes, ties, and node attributes that is very similar to the very popular ggplot2 R package. Figure 3 includes network diagrams, or sociograms, for a subsample of four classrooms included in the langnet.dta dataset. These diagrams illustrate the variety of classroom network contexts in this sample. Students at risk are indicated with a triangle, and students not at risk are indicated with a circle. We use options in the ggraph commands to create plots that would be suitable in most academic journals (i.e., black and white). But the default options will create plots that incorporate more color and are ideal for other presentation formats.

Statistical summaries of each classroom network can also be obtained via the igraph package. Network size, number of edges, density, reciprocity, and distance can be calculated using the vcount, ecoun, edge_density, reciprocity, and mean_distance functions, respectively. Table 4 provides a summary of all these

Table 4*Network-Level Summary Statistics by Classroom*

Class number	Network size	No. edges	Density	Reciprocity	Avg. distance
Class 101	22	49	0.11	0.24	2.76
Class 103	23	60	0.12	0.30	2.57
Class 105	16	42	0.18	0.33	2.55
Class 106	23	62	0.12	0.29	2.66
Class 201	22	42	0.09	0.29	2.89
Class 203	18	33	0.11	0.30	3.12
Class 204	22	50	0.11	0.20	3.25
Class 401	21	49	0.12	0.37	2.71
Class 403	21	53	0.13	0.27	3.15

Note. All classes were kindergarten classrooms. The leading digit of the class number refers to the school (1–4).

network-level descriptive statistics by classroom. Code found on the OSF page demonstrates how to calculate these estimates and then compile them into a table using tools from the tidyverse suite of packages. Summaries such as Table 4 are useful in identifying trends or outliers across networks and looking for classrooms that have levels of network density, reciprocity, and so forth, that are quite different from the others. In our case, we find the networks to be consistent in their properties (some of this is likely due to the prompt to students to identify their three best friends, not an open-ended number).

Step 3: Calculating Centrality for Each Student

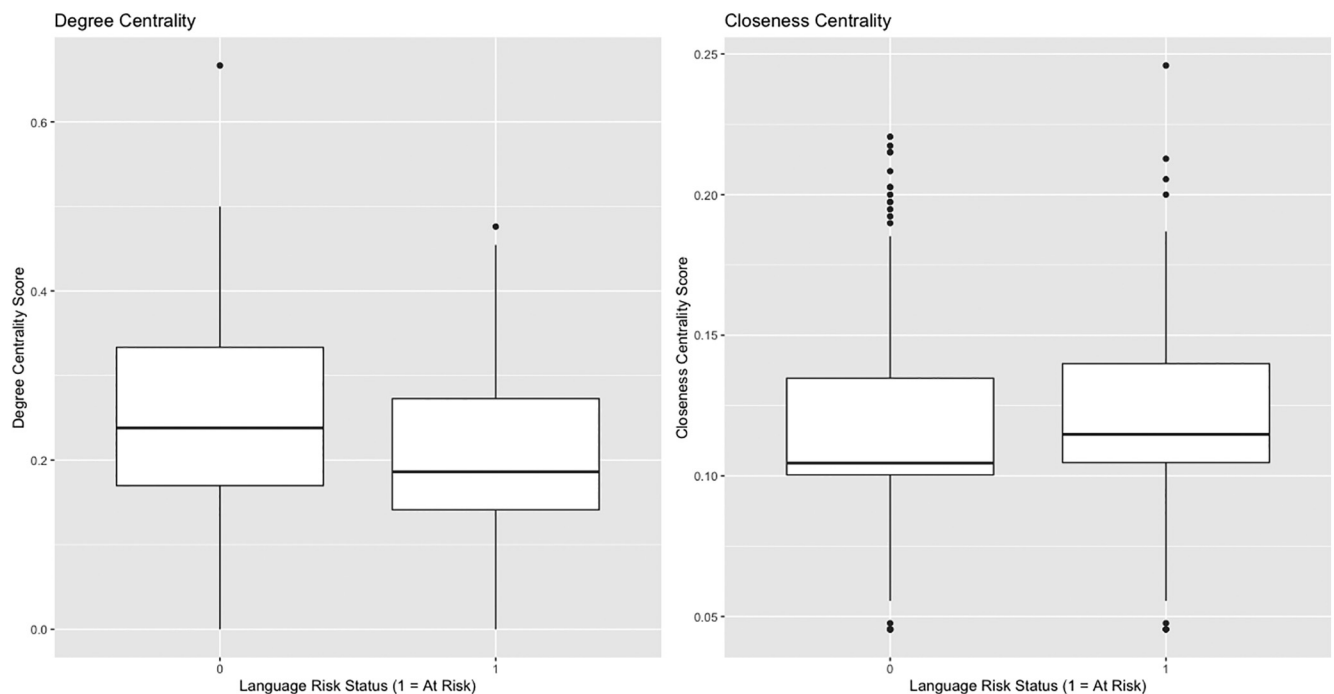
Having inputted the network data and summarized each network with sociograms and summary statistics, we next calculate the cen-

trality of each student in our dataset based on their position relative to their peers. As described above, many different measures of centrality can be used (igraph provides support for more than 40 unique measures of centrality); however, here we demonstrate the use of degree centrality and closeness. These two measures fit best with our research questions, which focus on how a student's language skills may be related to their perceived social position in the classroom. The degree and closeness functions in igraph will estimate those respective centrality measures for each student. We then take those scores and merge them into the larger dataset for statistical modeling.

It may be useful to provide a graphical summary such as a box plot or histogram to illustrate the distribution of centrality scores across the sample in order to begin to assess preliminary group differences. Figure 4 is one example of this, where we provide box

Figure 4

Comparing Distribution of Degree Centrality (Left) to Closeness Centrality (Right), by Language Risk Status



Note. $N = 188$.

plots for degree centrality and closeness side by side, separated by student risk status. From this plot, we can begin to infer that centrality may differ for those who are and are not at risk of a language disorder, although the direction of that difference may differ, with students at risk having somewhat lower degree centrality scores, but somewhat higher closeness.

Step 4: Modeling Centrality as a Function of Language Status

The final step in our investigation is to fit a series of statistical models that may help explain the relationship between student centrality (the outcome), language status (the predictor of interest), and other background and demographic characteristics (controls for common confounds). Multilevel modeling is often the preferred approach to this kind of question, as students are nested within classrooms, and this dependency is not captured by OLS regression models (Raudenbush & Bryk, 2002). However, in this case, we chose to fit our models using OLS regression, as the phenomenon in question, centrality, did not appear to be significantly clustered by classroom ($ICC = .08$). We also use bootstrap resampling to obtain more precise and unbiased standard errors for our model estimates. This is especially important, given that centrality measures can create a dependency structure in the data (i.e., students with higher degree centrality in a classroom necessitate the existence of other students in the classroom with higher centrality). We operationalized language in two ways: (a) using a binary variable that indicated whether a student was at risk of a language impairment (1 = yes, 0 = no); and (b) using the CELF score itself as a continuous measure of language skills. Therefore, here we run two sets of models, one with the CELF-5 cut score as a categorical risk variable, and one with the CELF-5 continuous score.

We first fit a main effects model which included language status as well as covariates including student gender, age, race/ethnicity, and classroom size. Depending on the nature of the research hypotheses, one might also fit an interaction model where the predictor of interest is interacted with other characteristics such as gender or age to see if the relationship between language and centrality is conditional upon these other covariates. See our OSF page for the code to implement these models as well as how to obtain bootstrapped confidence intervals for each estimate using the Boot function from the car package (Fox & Weisberg, 2019).

Full model results can be found in Table 5.³ In Model 1, we find that students above the recommended clinical threshold for risk of language disorders tend to also have lower degree centrality ($B = -.051$, 95% CI $[-.085, -.017]$) than their peers who did not score above the threshold, even after controlling for student age, gender, race/ethnicity, and classroom size. In Model 2, we find that students with higher CELF scores (suggesting higher language skills) tend to also have higher degree centrality ($B = .008$, 95% CI $[-.003, .013]$), again after controlling for the same covariates.

This set of findings demonstrates the utility of an SNA approach to answer novel empirical questions in psychology. To illustrate, it appears that language risk relates to lower social integration in a kindergarten classroom friendship network. This finding has important implications for research and practice due to the ubiquity of language screening and assessments upon school entry and can inform how teachers may use language screening to inform instruction related to social development and friendships. It also can inform systems and

Table 5
OLS Regression Estimates for the Relationship Between Language and Degree Centrality

Term	Model 1	Model 2
Intercept	-0.225 (0.210)	-0.209 (0.210)
Language risk status (1 = at risk)	-0.047** (0.017)	— —
CELF score	— —	0.008** (0.003)
Male	-0.029 (0.017)	-0.030 (0.017)
Age (years)	0.075*** (0.031)	0.065*** (0.031)
African American	-0.037 (0.023)	-0.035 (0.023)
Asian	-0.025 (0.098)	-0.021 (0.099)
Hispanic/Latinx	-0.072* (0.027)	-0.071* (0.027)
Other race/ethnicity	-0.006 (0.025)	-0.004 (0.025)

Note. $N = 188$ students. CELF = clinical evaluation of language fundamentals; OLS = ordinary least squares. Standard errors obtained via bootstrap resampling with 1,000 replications.

* $p < .05$. ** $p < .01$. *** $p < .001$.

policy level research on how schools and districts use data to inform decision making and subsequent instruction and intervention. However, it is important to note that these data were not longitudinal and preclude us from making inferences about the causal direction of finding. Future research using longitudinal data would be able to explore the contribution of language to social and friendship network development, as well as if friendships can support language learning and development. SNA provides a unique opportunity to examine behavioral and personality factors associated with who we choose as friends, and the extent to which those friends influence our beliefs, attitudes, and behavior (McPherson et al., 2001). Future work could use friendship and language data collected across the school year or multiple years to examine friendship selection and influence and language skills over time and combine other areas of interest that are likely to influence these developmental trajectories such as peer aggression, engagement, and relationships with teachers. Overall, identifying key factors that are associated with, and predictive of, friendships can help isolate important factors that contribute to student resiliency (e.g., strong friendships, strong communication skills) as well as targets for intervention.

As seen above, SNA offers a useful lens for more carefully unpacking the role of social context in classroom research. In this example, we can measure each student's social position in the classroom as a function of who they identify as close friends (and, conversely, who is identified by others as a close friend). This data then facilitates comparison of classroom networks, both visually and statistically, to examine patterns of association and friendship between classrooms. It also allows for the calculation of network centrality,

³ For the sake of brevity and clarity, we report only the results for models predicting degree centrality here. The OSF page found at <https://osf.io/abkcx/> includes the results of the models that used betweenness as an outcome.

which can then be used in a predictive model to examine the relationship between network position and measures of language ability.

Recommendations for Study Design When Researching Networks

Over the course of our work with social networks, we have learned several lessons that may be useful for others who are interested in applying this methodology. Thus, we have a few recommendations for future network researchers. First, the design of the network collector instrument should be closely informed by your research questions and conceptual framework. Often, we find that it is tempting to collect data on friends or close colleagues without thinking carefully about the ways in which one's closest supporters may in fact influence them via a network perspective. One way to think about this is to ask oneself what may be "flowing" through the network that relates most closely to the concept you are interested in—whether it is beliefs, attitudes, information, resources, and so forth. The possibilities are extensive and require the researcher to clearly articulate these relationships from the beginning. This does not preclude the team from also collecting more general information about close friends. This information might still be quite helpful. But it also is important for the prompt on the network collector to be specific enough to capture the kinds of information that connect most closely to your essential questions.

Second, as is the case in many other types of quantitative research, collecting longitudinal data is highly recommended when studying networks. Being able to compare network structures at two or more time points opens a wide range of additional hypotheses that can be pursued related to the relative persistence (or lack thereof) of subgroups and ties over time. This may be due to the impact of an intervention, or due to maturation or other social and developmental processes. Regardless, without capturing several waves of network data, one is limited in the types of conclusions that they can reach about the relative importance of the network in shaping individual beliefs, attitudes, or behaviors. Just as accounting for a pretest score can be useful in explaining variation in posttest scores, controlling for the state of one's "pre" network can add significant weight to conclusions about change over time.

Third, proactively designing network data collection to minimize missing data is critically important, particularly when using bounded networks. In classroom-based research, it is common to provide an incentive to the entire class (e.g., classroom pizza party) if 80% of student consent forms are returned (with caregiver permission to participate or a decline to participate). It is also important to communicate study goals to caregivers, detailing the types of questions students will be asked and how student responses to peer nominations will be shared with any school personnel and in research (e.g., at an aggregate level that will not identify individual student ties). Additionally, when recruiting administrators, teachers, caregivers, and students, we find it helpful to describe findings by Mayeux et al. (2007) that report minimal distress or discussion of social dynamics among students after participating in network related surveys. These decisions should be made in close collaboration with your Institutional Review Board, as we have come across variability in several institutions' approaches to network related data collection.

Finally, network data collection can be costly, particularly when including direct assessment data. A helpful approach to consider is to leverage collaborations and integrate network related questions into

ongoing studies. This approach can help minimize missing data and decrease project related costs (e.g., recruitment expenses) as well as provide opportunities to link traditional network data with resource-intensive developmental or intervention research measures.

Conclusion

We hope that this tutorial provides readers with an accessible introduction to how social network research methods might be applied to psychological research in applied settings such as schools, families, clinics, firms, or other settings where individuals are embedded in contexts that might shape their knowledge, beliefs, motivations, or attitudes. We have found these tools to be useful in helping disentangle the inherent complexities of students' experiences in classrooms, especially as those experiences relate to social interaction, learning, and other developmental processes.

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